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Review

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Vision-language foundation model for 3D medical imaging

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Jing Wu[1,7], Yuli Wang[2,7], Zhusi Zhong[3], Weihua Liao[4], Natalia Trayanova[5], Zhicheng Jiao[3] & Harrison X. Bai[2,6]

Recent advances in AI, especially vision-language foundation models (VLFMs), show promise in automating radiology report generation from complex 3D medical imaging data. Our review analyzes 23 studies on VLFMs, focusing on model architectures, capabilities, training datasets, and evaluation metrics. We discuss AI's evolution in radiology, emphasizing the need for diverse datasets and standardized metrics, as challenges remain in producing consistent, high-quality reports.

Radiology occupies a central role in contemporary healthcare, serving as a fundamental tool in the diagnosis, treatment planning, and monitoring of a myriad of diseases[1,2]. Among the advancements in medical imaging, 3D imaging technologies have revolutionized the field by providing unprecedented detail and depth, allowing for intricate visualizations of anatomical structures[3–5]. Despite these advancements, the complexity of interpreting 3D imaging data adds a significant burden on radiologists, who must integrate vast information swiftly to make accurate diagnostic decisions[6]. This growing complexity, compounded by an increasing volume of imaging studies and a global shortage of trained radiology professionals, creates a critical bottleneck in healthcare systems worldwide.

The advent of artificial intelligence (AI), particularly vision-language foundation models (VLFMs), has introduced the possibility of mitigating these challenges by automating the generation of radiological reports. These VLFMs are designed to synergistically combine the capabilities of advanced image processing algorithms and sophisticated natural language processing techniques[7,8]. By doing so, they aim to mimic the analytical processes of human radiologists, interpreting complex imaging data and articulating findings in the form of coherent, detailed reports.

This integration of visual and linguistic AI capabilities is particularly promising for handling the intricacies of 3D medical imaging. The models are not only tasked with recognizing and categorizing pathological features but also with describing these features and their implications for patient care in a clinically relevant and accessible manner. Despite the significant potential of these models to enhance diagnostic workflows, their practical application faces several challenges. Current AI technologies, while rapidly evolving, still struggle to consistently produce high-quality reports that can match the depth of analysis and insight provided by human experts. The need for extensive training data, the ability to generalize across diverse clinical scenarios, and the development of evaluation metrics that adequately capture the clinical utility and accuracy of AI-generated reports are among the foremost challenges.

This review paper aims to explore the recent advancements and the current state[8,9] of VLFMs specifically designed for 3D medical imaging. Figure 1 presents an overview of the VLFMs framework, encompassing key components such as datasets, model architecture, evaluation strategies, and the integration of VLFMs into clinical workflows. It assesses their capabilities, the challenges they face, and outlines future directions necessary for their integration into clinical practice. The structure of this paper is organized as follows: "Methods" section details our literature search strategy and specifies the inclusion criteria for studies on VLFMs considered in this review. "Findings and results" (i) introduces datasets used for VLFMs based on characteristics such as availability and size; (ii) investigates the technical foundations and architectural designs of these models; (iii) presents a summary of state-of-the-art evaluation metrics, including assessments by both human and machines; (iv) analyzes the quantitative performance of these models. "Discussion" highlights potential new research avenues, emphasizing the importance of incorporating additional datasets from various radiological modalities and the need for enhanced evaluation metrics as pivotal areas for future research and development. "Conclusion" provides a comprehensive summary of our perspectives on the current state of VLFMs.

Methods

Literature search strategy

A comprehensive literature search was conducted across several databases including PubMed, Google Scholar, Scopus and Arvix. The search was focused on identifying studies that describe the application of AI models, particularly VLFMs, for generating radiology reports and/or visual question answering (VQA) from 3D medical imaging data. Keywords used in the

[1]Department of Radiology, The Second Xiangya Hospital, Central South University, Changsha, Hunan, China. [2]Department of Radiology and Radiological Science, Johns Hopkins University School of Medicine, Baltimore, MD, USA. [3]Department of Diagnostic Imaging, Rhode Island Hospital, Providence, RI, USA. [4]Department of Radiology, Xiangya Hospital, Central South University, Changsha, Hunan, China. [5]Department of Biomedical Engineering, Johns Hopkins University, Baltimore, MD, USA. [6]Present address: Department of Radiology, University of Colorado, Aurora, CO, USA. [7]These authors contributed equally: Jing Wu, Yuli Wang.
e-mail: Harrison.bai@cuanschutz.edu

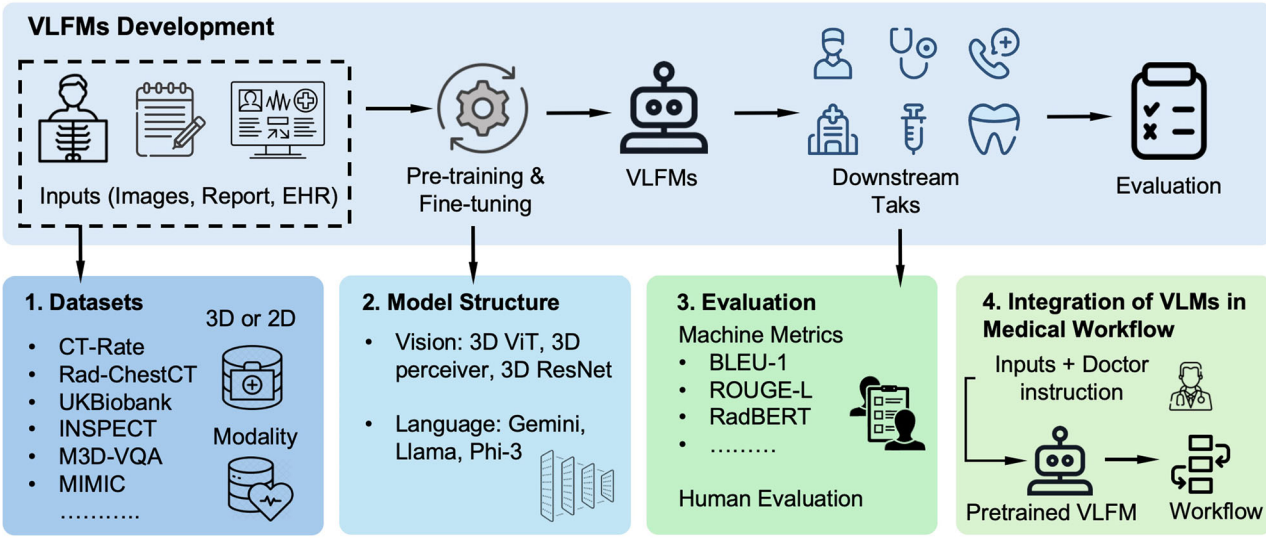

Fig. 1 | Overview of the VLFMs' framework. The training phase incorporates diverse input modalities—medical images, textual reports, metadata, and historical records—followed by large-scale pre-training[1]. Benchmark is performed across a wide range of medical datasets, such as CT-Rate, Rad-ChestCT, UKBiobank and others[2]. The framework employs advanced model structures including 3D Vision Transformer (ViT), 3D Perceiver, 3D ResNet etc. for the vision part and Gemini, Llama, Phi-3 etc. for the language model[3]. Evaluation includes both automated metrics—such as BLEU and ROUGE—and clinical-specific tools like the RadBERT, complemented by expert human evaluation[4]. VLFMs are integrated into clinical workflows, offering actionable insights that enhance diagnostic accuracy and support clinical decision-making.

search included combinations of "3D medical imaging", "AI in radiology", "vision-language models", "foundation model", "visual question answering", and "automated report generation".

Inclusion and exclusion criteria

We included studies that specifically addressed the development or application of VLFMs capable of interpreting 3D imaging data and generating corresponding radiological reports and/or performing VQA. Only studies that provided empirical evidence of the models' performance through recognized evaluation metrics were included. Conversely, studies were excluded if they did not focus on 3D imaging, were not original research (e.g., reviews or case report), or did not offer quantitative evaluation results. A comprehensive literature search was performed using electronic databases including PubMed, Google scholar, Scopus and Arxiv. The keywords included combinations of "CT" and/or "MRI" and/or "3D imaging"; "large language model" and/or "vision language model" and/or "foundation model"; "visual question answering" or "VQA"; and "report generation". Only studies published in English from January 2000 to May 2025 were included. From this search, 23 studies met all the inclusion criteria, underscoring the nascent stage of this research area. Summary of studies on report generation and/or VQA of 3D medical imaging using VLFMs is shown in Table 1. In the fast-paced field of AI-driven radiology, it's essential to balance peer-reviewed and non-peer-reviewed sources. Although peer-reviewed research remains the gold standard, valuable insights often surface through less formal channels, such as preprints and Arxiv. By including these sources in our review paper, we aim to highlight emerging trends and address unmet needs, offering a thorough perspective on both established and emerging research areas.

Findings and results

Datasets assessment

Data extracted from the selected studies included details on the AI model architecture, the evaluation metrics employed, and the reported model performance as shown in Table 2.

In the development and training of VLFMs for medical imaging, the availability and quality of datasets play a crucial role. While there are numerous datasets available for 2D imaging, such as the widely-used MIMIC dataset[10] for chest X-rays, the landscape for 3D medical imaging datasets is less robust. Foundation models (FMs)[11,12] often rely on image-caption pairs derived from publications on PubMed and various public datasets to enhance their learning capabilities. However, publicly available 3D datasets are significantly limited. To address this issue, a recent study introduced CT-RATE[13], a 3D medical imaging dataset that pairs images with textual reports. CT-RATE consists of 25,692 non-contrast chest CT volumes, expanded to 50,188 through various reconstructions, from 21,304 unique patients, along with corresponding radiology text reports. This dataset represents a valuable resource for training and validating VLFMs tailored specifically for chest CT. One limitation is that the reports have been translated from Turkish to English, potentially introducing nuances that affect the learning of medical terminology and report structuring. Furthermore, the availability of INSPECT[14], which includes 23,248 CT pulmonary angiography images as detailed in the paper, with 21,266 of these images publicly available, represents a significant advancement in diversifying data sources for advanced diagnostic AI tools. These developments underscore the need for more diverse and extensive 3D medical imaging datasets to further enhance the capabilities of AI-driven diagnostic technologies. Additionally, the M3D dataset[15] comprises 120 K image-text pairs and 662 K instruction-response pairs covering 8 tasks, including Image-Text Retrieval, Report Generation, Referring Expression Comprehension, Referring Expression Generation, Semantic Segmentation, and Referring Expression Segmentation. This multimodal dataset introduces richer dialogue corpus and three-dimensional segmentation tasks, highlighting important trends in current VLFM research.

Technical foundation for vision language models in 3D medical imaging

In the domain of medical imaging, particularly for the generation of radiology reports from 3D images, VLFMs play a crucial role by leveraging methodologies from computer vision and large language models (LLMs). As depicted in Fig. 2a, these models utilize a deep neural network architecture featuring distinct encoders for images and text. Here, 3D CNNs and Transformer-based networks extract volumetric features from 3D imaging data, while text encoders process input data tokens. These neural network architectures are essential for tasks such as lesion detection and organ segmentation, preserving the inherent spatial relationships inherent in 3D structures. For instance, RadGenome-Chest CT[16] is designed to train a

Table 1 | Summary of studies on report generation and/or VQA of 3D medical imaging using vision language foundation model

Paper	Model name	Dataset name	Dataset availability	Modality	Patients	E/R	Images	Language	Task
Hamamci et al.[13]	CT-CLIP	CT-RATE	Public	CT	21,304	25,692	50,188	English	VQA
		RAD-ChestCT			NA	3630	NA		
Bai et al.[15]	M3D-LaMed	M3D-Cap	Public	CT	NA	42,496	120,092	English	RG
		M3D-VQA				NA	96,170		VQA
Wu et al.[11]	RadFM	RP3D-Caption	Public	CT, MRI	NA	51k	51k	English	RG
		RP3D-VQA				142k	142k		VQA
Hamamci et al.[55]	CT2Rep	NA	NA	CT	21,314	25,701	25,701	NA	RG
	CT2RepLong				2807	7195	7195		
Chen et al.[56]	Dia-LLaMA	CTRG-Chest-548K	Public	CT	NA	1804	1804	Chinese and English	RG
Yang et al.[31]	Med-Gemini	CT-US1	Private	CT	NA	681,460	681,460	NA	RG
Blankemeier et al.[57]	Merlin	NA	Public	CT	18,321	25,528	25,528	English	RG
Zhang et al.[58]	NA	Prostate-MRI-USBiopsy dataset	Public	MRI	615	NA	NA	NA	VQA
		NA	Private		206				
Liu et al.[59]	HILT	CT-RATE	Public	CT	NA	24,284	NA	English	RG
		BIMCV-RG	Private			5328			
Zhong et al.[60]	Abn-BLIP	INSPECT	Public	CT	19,402	23,248	NA	English	RG
		BUH	Private		19,565	59,754			
Shui et al.[61]	fVLM	MedVL-CT69K	Private	CT	69,086	272,124	272,124	English	RG
Lee et al.[62]	MS-VLM	CT-RATE	Public	CT	21,304	25,692	50,188	English	RG, VQA
		In-house Rectal MRI dataset	Private	MRI	311	311	NA		
Ni et al.[63]	MG-3D	CTRG-Chest	Public	CT	NA	1804	1,804	English	RG
		CT-RATE				50,188	50,188		VQA
Chen et al.[64]	Reg2RG	RadGenome-ChestCT	Public	CT	21,304	25,692	25,692	English	RG
		CTRG-Chest-548K			NA	1804	1,804		
Lee et al.[65]	PIRTA	UKBiobank	Public	MRI	38,532	NA	NA	English	RG
		SNUH + SNUBH	Private		1045	1045			
		BRMH	Public		580	580			
		ISLES			206	206			
Shi et al.[66]	Med-2E3	M3D-Cap	Public	CT	NA	NA	117 K	English	RG
		M3D-VQA					866 K		VQA
Lai et al.[67]	E3D-GPT	BIMCV-R	Public	CT	NA	7518	NA	English	RG, VQA
		CT-RATE				50,188			
		Private Data	Private			300,360	NA		
Luo et al.[68]	VividMed	CT-RATE	Public	CT	NA	25,692	25,692	English	RG
Li et al.[69]	ViT3D Alignment of LLaMA3	AMOS-MM 2024 Dataset	Public	CT	NA	2088	2,088	English	RG, VQA
		CT-RATE			21,340	50,188	50,188		RG
Zheng et al.[70]	PCRL	CTRG-Brain	Public	CT	NA	6000	160,336	Chinese	RG
Chen et al.[71]	3D-CT-GPT	CT-RATE	Public	CT	NA	8070	8070	English	RG, VQA
		Dataset-XY	Private			1887	1887		
Alkhaldi et al.[72]	MiniGPT-Med	NLST	Public	CT	NA	7625	7625	English	RG, VQA
		MIMIC		CT, MRI		227,835	377,110		
Li et al.[73]	BrainGPT	3D-BrainCT	Private	CT	9689	18,885	NA	English	RG
		CQ500	Public		NA	133			

VQA visual question answering, *E/R* number of examinations and reports, *RG* report generation.

VLFM to generate reports based on both images and corresponding segmentations, enhancing interpretability with visual cues and improving model performance. Moving from 2D to 3D, VLFMs employ techniques such as multi-view learning[17] and volume rendering[18] to enhance the interpretation of complex anatomical structures and direct visualization of 3D volumes. During pre-training, as illustrated in Fig. 2b), these models use frameworks that facilitate robust interaction between image features and textual data. Contrastive learning[19] aligns 3D image features with textual

Table 2 | Detailed summary of the key aspects of published VLFMs, including architecture, evaluation metrics, and performance

Paper	Architecture (image)	Architecture (text)	NLG metrics	CE metrics	Performance summary
Hamamci et al.[13]	Encoder of CT-ViT, CXR-Bert text encoder	CLIP-based	NA (directly match)	AUROC, ACC, precision, and F1 scores	Multi-abnormality classification: AUROC: 0.643, F1 score: 0.663, ACC: 0.621, Precision: 0.353.
Bai et al.[15]	3D ViT, 3D perceiver	LLaMA, SegVol	BLEU, ROUGE, METEOR, BERT-Score, LLM Qwen-72B*	Recall	For RG, BLEU: 15.15, ROUGE: 0.196, METEOR: 0.144, BERT-Score: 0.885, LLM Qwen-72B*: 0.849. For VQA, recall: 0.990.
Wu et al.[11]	Visual encoder, perceiver module	LLM	BLEU, ROUGE, BERT-sim, UMLS_Precision, UMLS_Recall	ACC, F1 score	For RG, BLEU: 0.128, ROUGE: 0.182, UMLS Precision: 0.225, UMLS Recall: 0.121, BERT-Sim: 0.586. For VQA, BLEU: 0.306, ROUGE: 0.364, UMLS Precision: 0.318, UMLS Recall: 0.249, BERT-Sim: 0.678.
Hamamci et al.[55]	3D vision feature extractor, transformer encoder	Transformer decoder, multimodal transformer decoder	BLEU, METEOR, ROUGE-L	Precision, recall, F1 score	For RG, BLEU: 0.460, METEOR: 0.295, ROUGE-L: 0.459, Precision: 0.749, Recall: 0.548, F1 score: 0.534.
Chen et al.[56]	Pre-trained ViT3D, perceiver	LLaMA	BLEU, METEOR, ROUGE-L	Precision, recall, F1 score	For RG, BLEU: 0.512, METEOR: 0.263, ROUGE-L: 0.422, Precision: 0.421, Recall: 0.387, F1 score: 0.372.
Yang et al.[31]	Gemini 1.5 Pro vision encoder, Gemini video encoder	Transformer decoder	CIDEr, BLEU4, Rouge-L	ACC, RadGraph F1-score	The first large multimodal model-based RG for 3D CT volumes, with 53% of AI reports considered clinically acceptable.
Blankemeier et al.[57]	Image encoder (inflated 3D ResNet152)	CLIP-based decoder, RadLlama	RadGraph-F1, BERT Score, ROUGE-2, BLEU score	Recall	For RG, BLEU: 0.102, ROUGE-2: 0.262, BERT: 0.588, RadGraph-F1: 0.293.
Zhang et al.[58]	Image encoder, domain adapter layer	LLaMA	NA	ACC, MSE, MAE, precision, Recall, F1 metrics	The approach boosts the performance of current scoring techniques, with improvement of 0.048-0.070 in ACC.
Liu et al.[59]	ViT-Base	Llama3-8B-Instruct with LoRA adaptation	BLEU 1-4, METEOR, ROUGE 1-2, L, CIDEr	NA	HILT surpasses baselines in 3DHRG, scoring 0.461 (CT-RATE) and 0.410 (BIMCV-RG) across resolutions and zero-shot, using only 64 tokens efficiently.
Zhong et al.[60]	I3D ResNet152	Abn-QFormer	BLEU-1, BLEU-4, ROUGE-1, ROUGE-L, METEOR, BERT-F1	ACC, AUC, Sensitivity, Specificity, F1-score	Abn-BLIP excels in RG: BLEU-1 0.677, BLEU-4 0.112, ROUGE-L 0.873 comparing to state-of-the-art (SOTA).
Shui et al.[61]	ViT-base	BERT-base	ACC, GREEN, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDEr	AUC, ACC, Specificity, Sensitivity, F1-score, Precision	fVLM excels in RG: ACC 0.645, GREEN 0.402, BLEU-1 0.522, ROUGE-L 0.464.
Lee et al.[62]	2D ViT-B/16 (DINO pre-trained) + Z-former	Vicuna-7B-v1.5 (fine-tuned with LoRA)	BLEU-4, ROUGE-L, METEOR	Precision, Recall, F1-score	CT-RATE RG: BLEU-4: 0.232, ROUGE-L: 0.438, METEOR: 0.396, F1: 0.261; CT-RATE VQA: F1: 0.297; Rectal MRI RG: Qualitatively superior, no quantitative metrics provided.
Ni et al.[63]	3D Swin Transformer	RadBERT	BLEU-1, BLEU-2, BLEU-3, BLEU-4	AUC, Recall	CTRG-Chest RG: BLEU-1: 0.626, BLEU-2: 0.522, BLEU-3: 0.426, BLEU-4: 0.372; CT-RATE VQA: AUC: 0.770.
Chen et al.[64]	ViT3D, 3-layer ViT3D	LLaMA2-7B	BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L	Precision, Recall, F1	RadGenome-ChestCT RG: BLEU-1: 0.473, BLEU-2: 0.365, BLEU-3: 0.296, BLEU-4: 0.249, METEOR: 44.070, F1: 0.253; CTRG-Chest-548K Report Generation: BLEU-1: 49.630, BLEU-2: 41.430, BLEU-3: 35.910, BLEU-4: 32.040, METEOR: 49.710.
Lee et al.[65]	3D ViT	NA	NA	ACC@1 (Accuracy at k = 1)	PIRTA's ACC@1 is significantly higher than baselines (GPT-4o, LLaVA-Med), but exact numeric values were not provided in the text.
Shi et al.[66]	M3D-CLIP	Phi-3 (LLM)	BLEU@1, ROUGE@1, METEOR, BERT	ACC	RG: BLEU@1: 0.515, ROUGE@1: 0.545, BERT: 0.909; VQA: BLEU@1: 0.586, ACC: 0.824
Lai et al.[67]	3D ViT-Base + 3D Convolution Perceiver	Vicuna 7B (LLM) with LoRA	BLEU-1, ROUGE-1, METEOR, BERT-F1	Balanced Accuracy, Precision, Recall, F1-score	RG (CT-RATE): BLEU-1: 0.412, BERT-F1: 0.880; VQA (CT-RATE): BLEU-1: 0.520, BERT-F1: 0.938
Luo et al.[68]	ViT encoder	Vicuna-1.5-7B	BLEU-4, ROUGE-L, METEOR	Macro RadBERT F1, Micro RadBERT F1, Macro RadBERT FNR, Micro RadBERT FNR	RG:BLEU-4: 0.373, ROUGE-L: 0.373, METEOR: 0.312, Macro RadBERT F1: 0.312, Micro RadBERT F1: 0.395, Macro RadBERT FNR: 0.156, Micro RadBERT FNR: 0.149

Table 2 (continued) | Detailed summary of the key aspects of published VLFMs, including architecture, evaluation metrics, and performance

Paper	Architecture (image)	Architecture (text)	NLG metrics	CE metrics	Performance summary
Li et al.[69]	ViT3D image encoder from M3D-CLIP	Asclepius-Llama3-8B	Green Score	ACC	RG(AMOS-MM 2024): Green Score (Avg.): 0.300, Green Score (Chest): 0.200, Green Score (Abdomen): 0.320, Green Score (Pelvis): 0.370; VQA: ACC: 0.610.
Zheng et al.[70]	ResNet101	LLaMA3-8B	BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDEr	F1 Score	RG: BLEU-1: 0.620, BLEU-2: 0.547, BLEU-3: 0.494, BLEU-4: 0.453, METEOR: 0.331, ROUGE-L: 0.577, CIDEr: 0.964, F1 Score: 0.706.
Chen et al.[71]	CT-ViT (from CT-CLIP)	Vicuna-7B	BLEU, ROUGE-1, ROUGE-2, ROUGE-L, METEOR, BERTScore_F1	NA	RG: BLEU: 0.384, ROUGE-1: 0.475, ROUGE-L: 0.328, METEOR: 0.357, BERTScore_F1: 0.889.
Alkhaldi et al.[72]	EVA	LLaMA-2-chat	BERT-Sim, CheXbert-Sim	BERT-Sim(VQA), IoU	RG: BERT-Sim: 0.720, CheXbert-Sim: 0.301; VQA: BERT-Sim: 0.580.
Li et. Al.[73]	CLIP ViT-L/14	LLaMA-7B	BLEU-1, BLEU-4, METEOR, ROUGE-L, CIDEr-R	FORTE (F1-score for degree, landmark, feature, impression)	RG(internal testing): BLEU-1: 0.444, BLEU-4: 0.204, METEOR: 0.301, ROUGE-L: 0.476, CIDEr-R: 0.212; RG(external validation): ACC (midline shift): 0.910.

*Qwen-72B serves as an LLM-based evaluation tool within M3D-Bench, facilitating automated and precise assessment of model performance. *NLG* natural language generation, *CE* clinical efficacy, *LLM* large language model, *ACC* accuracy, *AUROC* area under the ROC curve, *BLEU* BiLingual Evaluation Understudy, *ROUGE* recall-oriented understudy for gisting evaluation, *BERT* bidirectional encoder representations from transformers, *METEOR* metric for evaluation of translation with explicit ordering, *RadGraph-F1* measures accuracy in extracting clinical entities from radiology reports, combining precision and recall into one score, *UMLS* unified medical language system, *UMLS Precision* evaluates the proportion of correctly identified medical entities matching standardized UMLS terms, *UMLS Recall* assesses completeness in capturing relevant medical entities from text, according to UMLS terms, *CIDEr* consensus-based image description evaluation, *MSE* mean squared error, *MAE* mean absolute error, *Green Score* evaluates the clinical accuracy of generated radiology reports by measuring overlap with expert-written reports.

descriptions by mapping two elements (e.g., an image patch and the corresponding section of the report) to similar points in a learned space, relative to dissimilar elements (negatives). In fine-tuning, depicted in Fig. 2 (c), techniques such as Region of Interest (RoI) Pooling adapted for 3D[20] focus on specific volumetric areas, while cross-modal fine-tuning[21] ensures that textual output closely aligns with 3D image features. The evaluation of these models in clinical settings often involves zero-shot and few-shot predictions to assess their ability to answer specific prompt questions (i.e., VQA) and/or generate radiology reports.

Evaluation of AI-generated radiology reports

The evaluation of VLFMs using both natural language generation (NLG) metrics and clinical efficacy metrics is critical for assessing the accuracy, coherence, and clinical relevance of AI-generated radiology reports (Fig. 3). NLG metrics like BiLingual Evaluation Understudy (BLEU)[22], recall-oriented understudy for gisting evaluation (ROUGE)[23], metric for evaluation of translation with explicit ordering (METEOR)[24], and consensus-based image description evaluation (CIDEr)[25] offer insights into linguistic alignment with traditionally authored reports. They assess various aspects from n-gram co-occurrence to semantic similarity, using advanced techniques such as synonym matching and contextual embedding analysis from BERT[26]. However, while robust for linguistic assessment, these metrics often do not fully capture the clinical utility of reports. They may overlook the depth of semantic and contextual understanding required for accurately interpreting complex 3D medical images.

To overcome these challenges, clinical efficacy metrics such as CheXpert/CheXbert[27] Labeling Scores and RadGraph F1[28] have been introduced. These tools assess the clinical accuracy of AI-generated reports by analyzing text to identify and evaluate specific clinical findings essential in 3D imaging, such as detailed anatomical structures and their spatial relationships. The Radiology Report Clinical Quality (RadCliQ[29]) metric integrates these methodologies to provide a holistic assessment of a report's quality, encompassing both linguistic precision and clinical accuracy.

Despite these advancements, the evaluation process faces challenges including inherent subjectivity and variability in human evaluations, the resource-intensive nature of expert reviews, and the complexities involved in error analysis of AI-generated reports. These evaluations are pivotal, as they entail radiologists assessing AI-generated reports against clinical benchmarks for readability, informational completeness, and accuracy, with a special focus on critical 3D contextual details. However, the current limitations of evaluation metrics highlight the need for ongoing development to better align AI model performance with the nuanced requirements of clinical practice in radiology, especially as it pertains to complex 3D medical imaging.

Performance of VLFM on 3D medical imaging

Several VLFMs have demonstrated notable performances, each employing unique approaches and datasets. The M3D-LaMed[15] model, distinguished by its integration of multi-modal LLMs for complex 3D image analysis, has shown superior performance in tasks such as image-text retrieval, report generation, and visual question answering. Trained on a variety of 3D medical images, M3D-LaMed leverages a pre-trained 3D vision encoder and a spatial pooling perceiver to effectively handle the intricate data structures inherent in 3D imaging. Rad-FM[11], utilizing a comprehensive radiology image and report database, improved clinical-specific metrics, achieving 22.49% for UMLS_Precision[30] and 12.07% for UMLS Recall. CT-CLIP[13] excelled in zero-shot multi-abnormality detection on CT scans, notably enhancing mean AUROC and F1 scores over state-of-the-art, fully supervised methods by 0.144 and 0.097 in internal validation and 0.099 and 0.074 in external validation, utilizing CT scans paired with electronic health records. Google's Med-Gemini[31], trained across multiple medical modalities, including radiology and genomics, reported that 53% of its AI-generated reports were clinically acceptable, though it acknowledged the need for advancements to reach expert-level quality. These models, with its advanced capabilities in handling 3D datasets, underscore significant

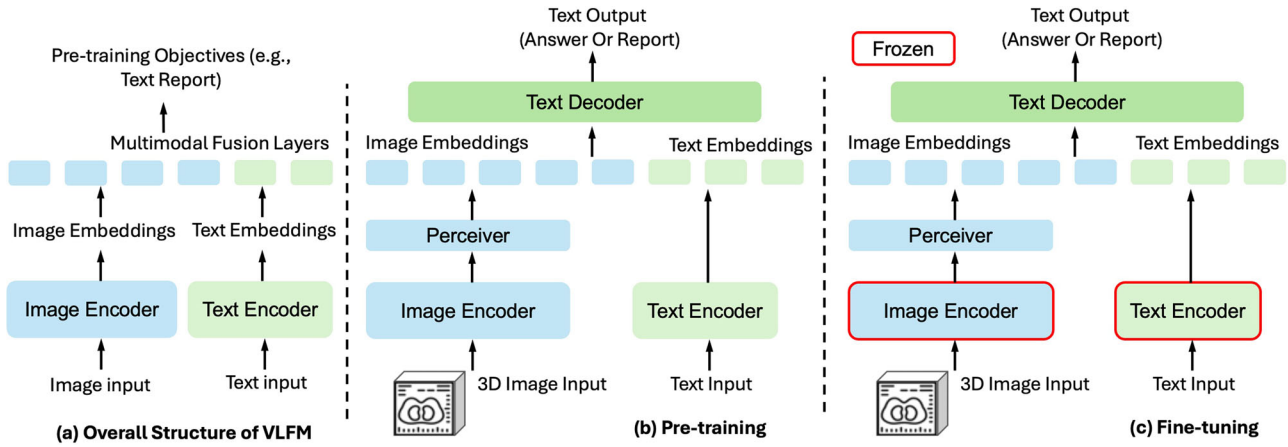

Fig. 2 | Overview of the VLFM Framework Across Pre-training and Fine-tuning Phases. Schematic diagram illustrating **a** the overall framework of VLFM, depicting its components during different phases: **b** pre-training strategy of VLFM and **c** fine-tuning strategy of VLFM. The diagram indicates that certain parts of the model may be frozen (not updated) during the fine-tuning phase. Typically, this would be the Image Encoder and Text Encoder to preserve the learned visual/text features. The Multimodal Perceiver is fine-tuned during both the pre-training and fine-tuning phases.

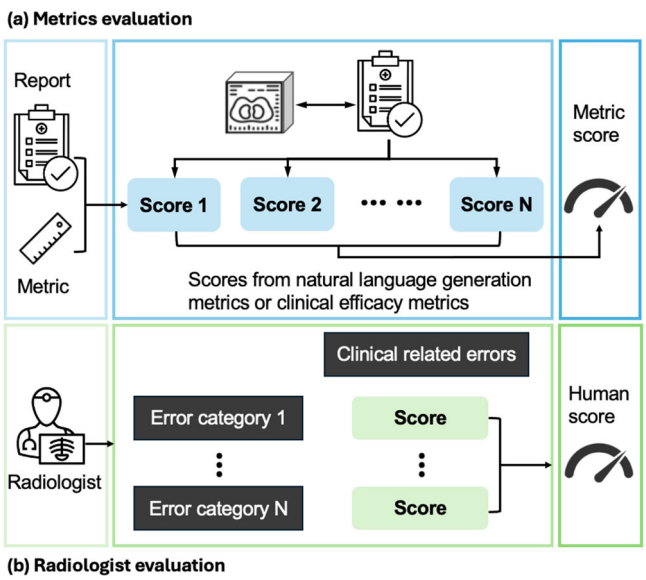

Fig. 3 | Comparison of Machine and Radiologist Evaluations of Radiology Reports. Comparisons between **a** machine metrics evaluation and **b** radiologist evaluation. In **a**, scores are calculated using specific metrics for a given radiology report. In **b**, radiologists evaluate the report by scoring clinical-related errors across various error categories.

advancements in applying AI to enhance the accuracy and utility of medical imaging report generation. They have the potential to improve the analytical capabilities of medical practitioners with detailed and reliable interpretations of complex 3D medical data.

Discussion

Challenges and future directions

Enhancement of high-quality and real-world datasets of image-reports pairs. A key challenge in advancing AI applications for 3D medical imaging, particularly for radiology report generation, is the lack of large, annotated datasets that encompass a wide range of pathologies across diverse patient populations. Comprehensive datasets are essential for training VLFMs that can accurately generate radiology reports from 3D images. Efforts to expand these datasets should focus on including a greater variety of 3D imaging types and ensuring detailed annotations that correlate imaging findings with clinical reports.

A significant limitation of current methodologies in radiology report generation from 3D medical images is their heavy reliance on metadata extracted from DICOM files. These metadata fields, typically only providing basic information about image modality and the body parts imaged, are inherently restrictive. This approach results in low-quality ground truth and often lacks the depth and context necessary for creating nuanced and clinically relevant reports. They fail to capture complex diagnostic information, critical nuances in pathology, and other subtleties essential for a comprehensive radiology report. Consequently, models trained on such datasets may develop a superficial understanding of the images, leading to generic and potentially inaccurate report generation.

This underscores the need for developing datasets that go beyond mere metadata to include rich, contextual annotations that directly relate specific imaging findings to detailed clinical insights. Collaborative initiatives with hospitals and research institutions to anonymize and share 3D imaging data could be vital in achieving this goal. The true potential of such collaborations can be realized through the establishment of large-scale, multi-modality imaging and report data consortiums. By pooling resources and datasets from diverse geographic and demographic sources, these consortia can create a more comprehensive and varied dataset that reflects a broader spectrum of pathologies, treatment outcomes, and patient populations.

This approach would not only enhance the volume and variety of data but also improve the robustness and generalizability of the AI models trained on them. Additionally, multi-site collaboration facilitates the standardization of data collection, annotation, and processing protocols, further enriching the quality of the data. Such an enriched dataset can serve as a cornerstone for developing more precise and contextually aware AI tools, ultimately leading to improved accuracy in medical imaging report generation and better patient care outcomes.

Domain-specific insights in medical imaging beyond general computer vision. To significantly improve the performance of VLFMs in generating accurate radiology reports from 3D medical images, it is crucial to focus on both the development and refinement of model architectures tailored to the inherent complexities of 3D medical scans[9]. The intricate spatial relationships and detailed anatomical structures present in these images necessitate the use of enhanced 3D convolutional layers, specifically designed to better capture the spatial hierarchies essential for accurately interpreting medical images.

Moreover, the integration of advanced language processing modules is indispensable. These modules must not only understand the clinical language but also articulate medical findings with high precision, effectively incorporating medical terminologies and nuanced patient data. Such

capabilities require a deep fusion of visual and textual understanding within the model architecture, ensuring that the generated reports are both medically accurate and contextually relevant.

Further augmenting the efficacy of these models, advanced training techniques like multi-task learning play a pivotal role. By enabling the model to simultaneously learn to identify specific medical conditions from 3D images and generate descriptive, clinically relevant text, multi-task learning enhances the model's ability to handle multiple tasks that mirror the workflow of human radiologists. This approach ensures a more holistic learning process, fostering models that are not only technically proficient but also practically applicable in clinical settings.

In addition to these architectural and training enhancements, the application of anatomical guidance tools such as TotalSegmentor can revolutionize model training[32]. By allowing precise segmentation of specific organs or regions within the 3D scans, these tools help create anatomical guidance in the image-text pair alignments. This guidance significantly aids the model in distinguishing between different anatomical features and their corresponding clinical descriptions, thereby refining the accuracy and relevance of the generated reports. Collectively, these strategies form a robust approach to overcoming current limitations and setting new benchmarks in the AI-driven generation of radiology reports from complex 3D medical imaging data.

Advanced metrics for assessing VLFMs in medical imaging report accuracy and clinical utility. Current metrics for evaluating VLFMs in medical imaging often fall short. These metrics, adapted from traditional NLP models, primarily measure textual similarity rather than clinical accuracy. This limitation can result in high scores for reports that appear textually accurate but miss critical diagnostic findings, impressions, and recommendations.

Although there are some efforts to include radiologists' subjective evaluations[33], these studies are limited to 2D chest X-rays and do not address the complexities of 3D imaging, which is more relevant to actual diagnostics and treatment. Evaluating VLFMs in 3D imaging requires more sophisticated metrics.

Studies[29,34–36] have highlighted the flaws in current metrics. BLEU struggles with identifying false findings, while BERTScore has higher errors in locating findings compared to CheXbert. These issues underscore the need for improved evaluation methods. The proposed RadCliQ[29] metric combines existing metrics with linear regression to create a more balanced evaluation model. However, RadCliQ's reliance on text overlap-based metrics and its testing on 2D datasets reveal limitations when applied to 3D imaging.

Future research should focus on developing metrics that accurately assess the clinical relevance of the generated reports, including diagnostic accuracy and terminology appropriateness. Advanced NLP techniques could also compare generated reports with a database of clinician-validated reports. By improving these metrics, researchers can establish more effective benchmarks for VLFMs in 3D medical imaging, ensuring the generated reports are not only accurate but also valuable in clinical practice, thereby enhancing patient care and diagnostic processes.

Controversy on simulated/synthesized 3D data for model training. The use of simulated or augmented 3D data in training VLFMs for medical applications presents significant challenges alongside its benefits. While it fills gaps in data availability, especially for rare or complex diagnostic scenarios, there is substantial debate about its reliability. Critics[37,33] argue that because simulated data do not originate from actual clinical experiences, they may not accurately reflect the complexities and variability necessary for training truly effective medical AI models.

Despite these concerns, advancements in generative AI offer promising solutions to enhance the reliability and quality control of synthesized data[38–40]. High-quality, AI-generated 3D datasets can now mimic real-world data with greater fidelity, reducing the gap between simulated and actual clinical scenarios. This evolution in data generation technology allows for an augmentation of existing 3D datasets, enhancing the breadth and depth of training environments for VLFMs without compromising the integrity of the models. By utilizing sophisticated generative techniques, including 3D-GAN[41], 3D diffusion probabilistic model[42], Neural Radiance Fields[43] and so on, developers can create augmented 3D datasets that are not only diverse but also closely aligned with real clinical conditions. These 3D datasets provide a controlled environment for testing and validating VLFMs, ensuring that the models are exposed to a wide range of medical scenarios, including rare and complex conditions. This approach not only enhances the model's diagnostic capabilities but also ensures that the VLFMs are robust and reliable, thereby potentially improving healthcare outcomes through more accurate and comprehensive medical imaging reports.

Addressing computational and clinical challenges in 3D transformer-based models. The integration of 3D transformer-based vision-language models in medical imaging represents a significant advancement, holding considerable promise for improved diagnostic accuracy and clinical reporting. However, the translation of these models from research prototypes to clinically deployable solutions face computational and clinical challenges that must be thoroughly addressed.

A primary obstacle to the widespread clinical adoption of traditional 3D transformers is their computational complexity. Specifically, these models suffer from quadratic memory and computational requirements associated with self-attention mechanisms, especially when processing high-resolution volumetric medical data[44,45]. Due to these constraints, practical implementations often necessitate significant compromises in input data resolution, thereby reducing the ability to capture fine-grained anatomical details essential for clinical diagnostics, such as subtle abnormalities, delicate vascular structures, nodules, and small lesions[46–48]. Consequently, these limitations directly impact the clinical applicability and diagnostic utility of transformer-based vision-language systems.

To overcome these computational challenges, recent developments in transformer architectures have introduced promising solutions. Hierarchical 3D swin transformers, for example, employ multi-scale hierarchical structures and locality-sensitive computations to substantially reduce memory usage without sacrificing model accuracy[49]. Additionally, sparse attention strategies, such as axial attention[50] and windowed attention mechanisms[51], have emerged as effective methods to selectively prioritize computations, significantly decreasing resource requirements while maintaining robust performance. Furthermore, hybrid CNN-transformer models exploit the complementary strengths of convolutional neural networks and transformers, combining efficient local feature extraction from CNNs with the global contextual understanding facilitated by transformers, thereby achieving a practical balance between computational efficiency and clinical efficacy[52,53]. Recent advances, such as the hierarchical attention approach proposed by Zhou et al.[54], exemplify these improvements by considerably reducing memory usage and computational demands. This strategy enables the analysis of higher-resolution volumetric data, resulting in improved performance in clinically relevant segmentation tasks, including pulmonary vessels and airways segmentation[54]. Such innovative solutions demonstrate significant progress toward resolving existing barriers, yet further advancements remain essential.

To address ongoing computational and clinical challenges comprehensively, future research should prioritize several key directions. Enhanced hierarchical attention methods capable of dynamically allocating computational resources based on clinical significance represent a promising area of exploration. Additionally, the development of transformer architectures specifically tailored for medical imaging, incorporating domain knowledge to optimize performance and efficiency, remains a critical research priority. Finally, investigating memory-efficient training paradigms, including model distillation, quantization, and efficient training strategies, will be crucial to improving practical feasibility. Explicitly recognizing and systematically addressing these computational and clinical limitations in this review aims to provide valuable insights and actionable guidance for

Table 3 | Summary of limitations of 2D vision-language foundation models (VLFMs), example studies, and clinical impact

Limitation category	Example studies	Impact on clinical adaptation
Interpretability and Clinical Trust	Rajpurkar et al.[74]; Rudin et al.[75]	Reduced trust; Poor interpretability; Reluctance to integrate models into clinical practice.
Domain Generalization and Robustness	Alijani et al.[76]; Manzari et al.[77]	Limited reliability across diverse clinical environments.
Computational Efficiency and Feasibility	Ghosh et al.[78]; Awais et al.[79]	Limited practical feasibility; integration barriers in clinical workflows.
Regulatory and Ethical Considerations	Mennella et al.[80]	Increased scrutiny; slower adoption due to regulatory and ethical barriers.

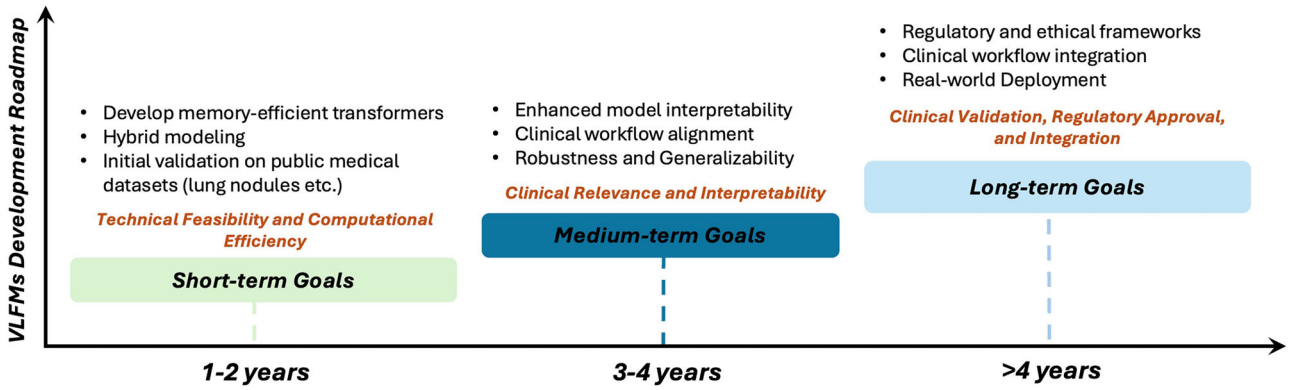

Fig. 4 | Future Roadmap for Clinical Integration of 3D VLFMs. The practical roadmap for future research and clinical integration of 3D VLFMs, including the short-term, medium-term and long-tern goals.

researchers committed to developing practical, efficient, and clinically impactful 3D transformer-based vision-language models in radiology.

Lessons learned from 2D VLFMs and a roadmap for future research on 3D VLFMs. The development and deployment of VLFMs in clinical environments represent promising yet challenging objectives. Although 2D VLFMs have exhibited encouraging results in controlled research environments, their real-world applicability and acceptance in clinical workflows have thus far remained limited. Understanding these limitations offers valuable insights that can guide the development and eventual clinical translation of more complex 3D VLFMs.

Several critical insights have been identified from examining the limitations inherent to existing 2D VLFMs. First, *interpretability and clinical trust* present significant challenges. Despite high quantitative performance demonstrated in research studies, clinicians frequently express reservations regarding the interpretability and transparency of these models' decisions, underscoring the necessity of incorporating explainability methods, uncertainty quantification, and clear visual justifications into model predictions. Second, the issue of *domain generalization and robustness* is a crucial barrier. Models trained on datasets from specific institutions or limited imaging protocols often struggle to generalize effectively to diverse clinical environments. This limitation highlights the importance of robust training methodologies, domain adaptation techniques, and comprehensive validation on heterogeneous datasets. Third, *computational efficiency and practical feasibility* remain pivotal for real-world deployment. Despite being simpler than 3D models, current 2D VLFMs still encounter challenges related to computational demands, limiting their integration into clinical workflows, especially in resource-constrained settings. Lastly, *regulatory approval and ethical considerations* significantly impact clinical adoption. Factors such as patient privacy, data security, and algorithmic biases must be comprehensively addressed to facilitate the real-world integration of these models into healthcare practice. These lessons and their corresponding implications for clinical adoption are summarized in Table 3.

Based on these insights from 2D VLFMs, we propose the following practical roadmap for future research and clinical integration of 3D VLFMs, which is shown in Fig. 4. Explicitly integrating these considerations into the research agenda will significantly enhance the likelihood of translating 3D VLFMs into clinically valuable tools, ultimately improving patient care through more precise and insightful medical image interpretation.

In the short-term (1–2 years), efforts should prioritize establishing technical feasibility and computational efficiency. This involves reducing computational complexity by developing memory-efficient transformer architectures, such as hierarchical (e.g., Swin), axial, or sparse attention mechanisms, specifically tailored for medical 3D imaging. Additionally, exploring hybrid modeling strategies that integrate transformer models with convolutional neural networks will help balance interpretability, accuracy, and computational demands. Initial validations can be conducted using standard public medical imaging datasets, focusing on benchmarking tasks such as fine-grained abnormality detection and segmentation (e.g., lung nodules, brain tumors). The mid-term goals (2–4 years) should focus on enhancing clinical relevance and interpretability. This phase includes integrating transformer models with advanced explainability techniques, including attention visualization and feature attribution methods. Furthermore, close collaboration with clinical experts is essential to clearly define realistic clinical scenarios such as preliminary report generation, screening assistance, and triage prioritization. Robustness and generalizability of models must be thoroughly validated through multi-institutional studies that encompass diverse populations and imaging modalities, alongside establishing standardized reporting criteria for consistent evaluation. In the long-term (4–7 years), extensive prospective clinical validation studies are essential, involving large-scale, multicenter clinical trials to rigorously demonstrate the clinical effectiveness, safety, and value of these models. Concurrently, proactive engagement with regulatory bodies (e.g., FDA, CE marking) is crucial to ensure compliance with ethical standards, transparency, fairness, bias mitigation, and reproducibility. Finally, successful integration into clinical workflows requires the development of interoperability standards, facilitating seamless integration with existing healthcare systems such as PACS, RIS, and EMR. Post-deployment, continuous monitoring and evaluation must be maintained to ensure sustained clinical benefit and system reliability.

Overall, the primary aimed scenarios for VLFMs focus on realistic, practical applications rather than full automation of medical diagnosis. Specifically, VLFMs are well-suited for supportive roles such as preliminary anomaly detection and triage assistance to streamline clinical workflows, as

well as assisting radiologists in generating structured clinical reports by automating routine descriptive tasks. Additionally, patient-focused applications involve providing simplified, patient-friendly explanations of imaging results, enhancing patient understanding and health literacy. Emphasizing interpretability, robust validation, clinician collaboration, and integration into existing clinical workflows will ensure these models have meaningful clinical impact and broad acceptance.

Conclusion

Improving VLFMs for radiology report generation from 3D medical imaging will have profound implications for the field of radiology, enhancing the speed and accuracy of diagnostic processes. These advancements will not only optimize the workflow within medical imaging departments but also potentially improve patient outcomes by providing faster and more accurate diagnostic information. As these AI systems become more integrated into clinical practice, they will play a crucial role in supporting radiologists by providing reliable preliminary reports that can be quickly reviewed and adjusted, if necessary, thereby streamlining the diagnostic workflow and allowing radiologists to focus on more complex cases. Moreover, the advancements in developing a vision-language foundation model specifically for 3D medical imaging not only revolutionize radiological practices but also hold profound implications for the broader field of medicine, such as pathology and radiation oncology. By integrating and interpreting complex imaging data more effectively, this technology can enhance diagnostic accuracy across various specialties, facilitate personalized treatment plans, and ultimately improve patient outcomes.

Data availability

No datasets were generated or analysed during the current study.

Abbreviations

AI	artificial intelligence
VLFMs	vision-language foundation models
VQA	visual question answering
FMs	Foundation models
RoI	Region of Interest
NLP	Natural Language Processing
CNN	Convolutional Neural Network
NLG	Natural Language Generation
CE	Clinical Efficacy
LLM	Large Language Model
ACC	Accuracy
AUROC	Area under the ROC curve
BLEU	BiLingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
BERT	Bidirectional Encoder Representations from Transformers
METEOR	Metric for Evaluation of Translation with Explicit Ordering
UMLS	Unified Medical Language System
CIDEr	Consensus-based Image Description Evaluation
MSE	Mean Squared Error
MAE	Mean Absolute Error
RadCliQ	Radiology Report Clinical Quality
ViT	Vision Transformer

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Author contributions

J.W. and Y.W. contributed equally to this work as co-first authors. They were responsible for the initial literature search, critical evaluation of the sources, and drafting of the initial manuscript. Z.Z. and Z.J. provided substantial contributions to the conceptualization of the review and the analysis of the literature. W.L. and N.T. contributed to the clinical perspective of the review and offered critical feedback on the intellectual content of the review. H.B. served as the corresponding author, overseeing all stages of the review process, including the final approval of the version to be submitted. All authors have read and approved the final version of the review.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Harrison X. Bai.

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